

Universidad Nacional de Loja  
 Facultad de Energía, Industria y Recursos Naturales No Renovables  
 Carrera de Computación  
 UNIDAD 2 - APE 2 - FASE 1

Nombre: Yuleisy Cecibel Saramillo Saramillo

Fecha: 01/06/2026

Asignatura: Matemáticas Discretas

Primer Ejercicio

$$ABC + A\bar{B}C + \bar{A}BC + A\bar{B}\bar{C} + \bar{A}\bar{B}C$$

$$ABC + AB\bar{C} + \bar{A}BC + \bar{A}\bar{B}C + A\bar{B}C$$

$$AB(C + \bar{C}) + \bar{A}BC + \bar{A}\bar{B}C + A\bar{B}C$$

$$AB(C \cdot 1) + \bar{A}BC + \bar{A}\bar{B}C + A\bar{B}C$$

$$AB + \bar{A}BC + \bar{A}\bar{B}C + A\bar{B}C$$

$$AB + \bar{A}C(B + \bar{B}) + A\bar{B}C$$

$$AB + \bar{A}C(C \cdot 1) + A\bar{B}C$$

$$AB + AC + A\bar{B}C$$

$$AB + A\bar{B}C + \bar{A}C$$

$$A(B + \bar{B}C) + \bar{A}C$$

$$A(B + C) + \bar{A}C$$

$$AB + AC + \bar{A}C$$

$$AB + C(A + \bar{A})$$

$$AB + C \cdot 1$$

$$AB + C$$

$$A \cdot B \cdot C + A \cdot \bar{B} \cdot C + \bar{A} \cdot B \cdot C + A \cdot B \cdot \bar{C} + \bar{A} \cdot \bar{B} \cdot C$$

|   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 0 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 0 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 1 | 1 | 1 | 0 | 0 |
| 0 | 0 | 0 | 0 | 1 | 0 | 0 | 0 | 1 | 0 | 1 | 0 | 1 | 0 | 0 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 1 |
| 0 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 1 | 1 | 0 | 0 | 0 | 0 | 1 | 0 | 1 | 0 | 1 | 0 | 0 | 0 | 0 |
| 0 | 0 | 1 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 0 | 0 | 1 | 0 | 0 | 1 | 0 | 0 | 0 |
| 1 | 0 | 0 | 0 | 0 | 0 | 1 | 1 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 0 | 0 | 1 | 0 | 0 | 1 | 0 | 0 |
| 1 | 0 | 0 | 0 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 0 | 0 | 0 | 0 | 1 | 1 | 0 | 0 | 0 | 0 | 1 | 0 | 0 | 1 | 0 |
| 1 | 1 | 1 | 0 | 0 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 0 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 0 | 0 | 0 |
| 1 | 1 | 1 | 1 | 1 | 1 | 1 | 0 | 0 | 0 | 1 | 1 | 0 | 0 | 1 | 0 | 1 | 1 | 1 | 1 | 0 | 0 | 1 | 0 | 0 | 0 | 1 |

$$A \cdot B + C$$

|   |   |   |   |   |
|---|---|---|---|---|
| 0 | 0 | 0 | 0 | 0 |
| 0 | 0 | 0 | 1 | 1 |
| 0 | 0 | 1 | 0 | 0 |
| 0 | 0 | 1 | 1 | 1 |
| 1 | 0 | 0 | 0 | 0 |
| 1 | 0 | 0 | 1 | 1 |
| 1 | 1 | 1 | 1 | 0 |
| 1 | 1 | 1 | 1 | 1 |



## Segundo Ejercicio

$$\begin{aligned}
 & ABCD + ABC\bar{D} + \bar{A}BCD + A\bar{B}\bar{C}D \\
 & ABC(D + \bar{D}) + \bar{A}BCD + A\bar{B}\bar{C}D \\
 & ABC(1) + \bar{A}BCD + A\bar{B}\bar{C}D \\
 & ABC + \bar{A}BCD + A\bar{B}\bar{C}D \\
 & BC(A + \bar{A}D) + A\bar{B}\bar{C}D \\
 & BC(A + D) + A\bar{B}\bar{C}D \\
 & ABC + BCD + A\bar{B}\bar{C}D
 \end{aligned}$$

$$ABCD + ABC\bar{D} + \bar{A}BCD + A\bar{B}\bar{C}D \quad \text{su simplificación es } ABC + BCD + A\bar{B}\bar{C}D$$

| A · B · C · D |   |   |   | + | A · B · C · $\bar{D}$ |   |   |   | + | $\bar{A}$ · B · C · D |   |   |   | + | A · $\bar{B}$ · $\bar{C}$ · D |   |   |   |
|---------------|---|---|---|---|-----------------------|---|---|---|---|-----------------------|---|---|---|---|-------------------------------|---|---|---|
| 0             | 0 | 0 | 0 | 0 | 0                     | 0 | 0 | 0 | 0 | 0                     | 0 | 0 | 0 | 0 | 0                             | 0 | 0 | 0 |
| 0             | 0 | 0 | 0 | 1 | 0                     | 0 | 0 | 0 | 0 | 0                     | 0 | 1 | 0 | 0 | 0                             | 1 | 0 | 0 |
| 0             | 0 | 0 | 1 | 0 | 0                     | 0 | 0 | 0 | 0 | 0                     | 1 | 0 | 0 | 0 | 1                             | 0 | 1 | 0 |
| 0             | 0 | 0 | 1 | 1 | 0                     | 0 | 0 | 0 | 0 | 1                     | 0 | 0 | 0 | 1 | 0                             | 0 | 0 | 0 |
| 0             | 0 | 1 | 0 | 0 | 0                     | 0 | 0 | 0 | 0 | 1                     | 0 | 0 | 0 | 1 | 0                             | 1 | 0 | 0 |
| 0             | 0 | 1 | 0 | 1 | 0                     | 0 | 0 | 0 | 0 | 1                     | 1 | 0 | 0 | 0 | 0                             | 0 | 0 | 1 |
| 0             | 0 | 1 | 1 | 0 | 0                     | 0 | 0 | 0 | 0 | 1                     | 1 | 0 | 0 | 0 | 0                             | 0 | 1 | 0 |
| 0             | 0 | 1 | 1 | 1 | 0                     | 0 | 0 | 0 | 0 | 1                     | 1 | 1 | 0 | 0 | 0                             | 0 | 0 | 0 |
| 1             | 0 | 0 | 0 | 0 | 0                     | 0 | 0 | 0 | 0 | 1                     | 0 | 0 | 0 | 0 | 0                             | 0 | 0 | 1 |
| 1             | 0 | 0 | 0 | 1 | 0                     | 0 | 0 | 0 | 0 | 0                     | 0 | 0 | 0 | 0 | 1                             | 1 | 1 | 1 |
| 1             | 0 | 0 | 1 | 0 | 0                     | 0 | 0 | 0 | 0 | 0                     | 0 | 0 | 0 | 0 | 1                             | 1 | 0 | 0 |
| 1             | 0 | 0 | 1 | 1 | 0                     | 0 | 0 | 0 | 0 | 0                     | 0 | 0 | 0 | 0 | 1                             | 1 | 0 | 0 |
| 1             | 1 | 0 | 0 | 0 | 0                     | 0 | 0 | 0 | 0 | 0                     | 0 | 0 | 0 | 0 | 0                             | 0 | 0 | 1 |
| 1             | 1 | 0 | 0 | 1 | 0                     | 0 | 0 | 0 | 0 | 0                     | 0 | 0 | 0 | 0 | 0                             | 0 | 0 | 0 |
| 1             | 1 | 1 | 0 | 0 | 0                     | 0 | 0 | 0 | 0 | 0                     | 0 | 0 | 0 | 0 | 0                             | 0 | 0 | 0 |
| 1             | 1 | 1 | 1 | 0 | 0                     | 0 | 0 | 0 | 0 | 0                     | 0 | 0 | 0 | 0 | 0                             | 0 | 0 | 0 |
| 1             | 1 | 1 | 1 | 1 | 0                     | 0 | 0 | 0 | 0 | 0                     | 0 | 0 | 0 | 0 | 0                             | 0 | 0 | 0 |

| A · B · C |   |   | + | B · C · D |   |   | + | A · $\bar{B}$ · $\bar{C}$ · D |   |   |
|-----------|---|---|---|-----------|---|---|---|-------------------------------|---|---|
| 0         | 0 | 0 | 0 | 0         | 0 | 0 | 0 | 0                             | 1 | 0 |
| 0         | 0 | 0 | 0 | 0         | 0 | 1 | 0 | 0                             | 1 | 0 |
| 0         | 0 | 0 | 1 | 0         | 0 | 0 | 0 | 0                             | 1 | 0 |
| 0         | 0 | 1 | 0 | 0         | 0 | 0 | 0 | 0                             | 1 | 0 |
| 0         | 0 | 1 | 0 | 0         | 0 | 1 | 0 | 0                             | 0 | 0 |
| 0         | 0 | 1 | 1 | 0         | 0 | 0 | 0 | 0                             | 0 | 0 |
| 1         | 0 | 0 | 0 | 0         | 0 | 0 | 0 | 1                             | 1 | 1 |
| 1         | 0 | 0 | 0 | 0         | 1 | 0 | 0 | 0                             | 0 | 0 |
| 1         | 0 | 0 | 1 | 0         | 0 | 0 | 0 | 1                             | 1 | 0 |
| 1         | 0 | 1 | 0 | 0         | 0 | 0 | 0 | 1                             | 1 | 0 |
| 1         | 1 | 0 | 0 | 0         | 0 | 0 | 0 | 1                             | 0 | 0 |
| 1         | 1 | 1 | 0 | 0         | 0 | 1 | 0 | 0                             | 0 | 1 |
| 1         | 1 | 1 | 1 | 0         | 0 | 1 | 0 | 0                             | 0 | 0 |
| 1         | 1 | 1 | 1 | 1         | 0 | 0 | 0 | 0                             | 0 | 0 |



### Tercer Ejercicio

$AB\bar{C}D + A\bar{B}C + AB\bar{C}\bar{D} + A\bar{B}\bar{C} + A\bar{B}\bar{C}D + AC + A\bar{B}\bar{C}$   
 $A\bar{B}C + AC + AB\bar{C}D + AB\bar{C}\bar{D} + A\bar{B}\bar{C} + A\bar{B}\bar{C} + A\bar{B}\bar{C}D$   
 $AC(\bar{B} + 1) + AB\bar{C}D + A\bar{B}\bar{C}\bar{D} + A\bar{B}\bar{C} + A\bar{B}\bar{C} + A\bar{B}\bar{C}D$   
 $AC(1) + AB\bar{C}D + AB\bar{C}\bar{D} + A\bar{B}\bar{C} + A\bar{B}\bar{C} + A\bar{B}\bar{C}D$   
 $AC + AB\bar{C}D + AB\bar{C}\bar{D} + A\bar{B}\bar{C} + A\bar{B}\bar{C} + A\bar{B}\bar{C}D$   
 $AC + AB\bar{C}(D + \bar{D}) + A\bar{B}\bar{C} + A\bar{B}\bar{C} + A\bar{B}\bar{C}D$   
 $AC + AB\bar{C}(1) + A\bar{B}\bar{C} + A\bar{B}\bar{C} + A\bar{B}\bar{C}D$   
 $AC + AB\bar{C} + A\bar{B}\bar{C} + A\bar{B}\bar{C} + A\bar{B}\bar{C}D$   
 $AC + AB\bar{C} + \bar{B}\bar{C}(A + \bar{A}) + A\bar{B}\bar{C}D$   
 $AC + AB\bar{C} + \bar{B}\bar{C}(1) + A\bar{B}\bar{C}D$   
 $AC + AB\bar{C} + \bar{B}\bar{C} + A\bar{B}\bar{C}D$   
 $A(C + \bar{B}\bar{C}) + \bar{B}\bar{C} + A\bar{B}\bar{C}D$   
 $A(C + B) + \bar{B}\bar{C} + A\bar{B}\bar{C}D$   
 $AC + AB + \bar{B}\bar{C} + A\bar{B}\bar{C}D$   
 $AB + A\bar{B}\bar{C}D + AC + \bar{B}\bar{C}$   
 $B(A + \bar{A}\bar{C}D) + AC + \bar{B}\bar{C}$   
 $B(A + \bar{C}D) + AC + \bar{B}\bar{C}$   
 $AB + B\bar{C}D + AC + \bar{B}\bar{C}$   
 $B\bar{C}D + \bar{B}\bar{C} + AB + AC$   
 $\bar{C}(BD + \bar{B}) + AB + AC$   
 $\bar{C}(\bar{B} + D) + AB + AC$   
 $\bar{B}\bar{C} + \bar{C}D + AB + AC$   
 $\bar{B}\bar{C} + \bar{C}D + A(B + C)$   
 $\bar{C}(\bar{B} + D) + A(B + C)$

| $A \cdot B \cdot \bar{C} \cdot D$ | $A \cdot \bar{B} \cdot C$ | $A \cdot B \cdot \bar{C} \cdot \bar{D}$ | $A \cdot \bar{B} \cdot \bar{C}$ | $\bar{A} \cdot B \cdot \bar{C} \cdot D$ | $A \cdot C$ | $\bar{A} \cdot \bar{B} \cdot \bar{C}$ |
|-----------------------------------|---------------------------|-----------------------------------------|---------------------------------|-----------------------------------------|-------------|---------------------------------------|
| 0000100                           | 000100                    | 000001010                               | 001010                          | 10001000000                             | 111111      | 111111                                |
| 0000101                           | 000100                    | 000001000                               | 001010                          | 10001010000                             | 111111      | 111111                                |
| 0000000                           | 000101                    | 000000010                               | 001000                          | 10000000010                             | 111111      | 111111                                |
| 0000001                           | 000101                    | 000000000                               | 001000                          | 10000010010                             | 111111      | 111111                                |
| 0010100                           | 000000                    | 000100010                               | 000000                          | 11111000000                             | 111111      | 111111                                |
| 0010101                           | 000000                    | 000100000                               | 000001                          | 11111111100                             | 111111      | 111111                                |
| 0010000                           | 000001                    | 000100010                               | 000000                          | 11100000001                             | 111111      | 111111                                |
| 0010001                           | 000001                    | 000100000                               | 000000                          | 11100010001                             | 111111      | 111111                                |
| 1000100                           | 011100                    | 010001011                               | 111111                          | 100001001100                            | 111111      | 111111                                |
| 1000101                           | 011100                    | 010001001                               | 111111                          | 100001011100                            | 111111      | 111111                                |
| 1000000                           | 111111                    | 110000011                               | 111100                          | 100000001111                            | 111111      | 111111                                |
| 000001                            | 111111                    | 110000001                               | 111100                          | 100000011111                            | 111111      | 111111                                |
| 111100                            | 010000                    | 1111111111                              | 10001                           | 100101001100                            | 111111      | 111111                                |
| 111111                            | 110000                    | 1111111001                              | 10001                           | 100101011100                            | 111111      | 111111                                |
| 110000                            | 010001                    | 011100010                               | 10000                           | 000100001111                            | 111111      | 111111                                |
| 110001                            | 010001                    | 011100000                               | 10000                           | 000100011111                            | 111111      | 111111                                |



$$\bar{C} \cdot (\bar{B} + D) + A \cdot (B + C)$$

|     |       |       |       |
|-----|-------|-------|-------|
| 1 1 | 1 1 0 | 1 0 0 | 0 0 0 |
| 1 1 | 1 1 1 | 1 0 0 | 0 0 0 |
| 0 0 | 1 1 0 | 0 0 0 | 0 1 1 |
| 0 0 | 1 1 1 | 0 0 0 | 0 1 1 |
| 1 0 | 0 0 0 | 0 0 0 | 1 1 0 |
| 1 1 | 0 1 1 | 1 0 0 | 1 1 0 |
| 0 0 | 0 0 0 | 0 0 0 | 1 1 1 |
| 0 0 | 0 1 1 | 0 0 0 | 1 1 1 |
| 1 1 | 1 1 0 | 1 1 0 | 0 0 0 |
| 1 1 | 1 1 1 | 1 1 0 | 0 0 0 |
| 0 0 | 1 1 0 | 1 1 1 | 0 1 1 |
| 0 0 | 1 1 1 | 1 1 1 | 0 1 1 |
| 1 0 | 0 0 0 | 1 1 1 | 1 1 0 |
| 1 1 | 0 1 1 | 1 1 1 | 1 1 0 |
| 0 0 | 0 0 0 | 1 1 1 | 1 1 1 |
| 0 0 | 0 1 1 | 1 1 1 | 1 1 1 |

Ejercicio 4º

$$(\bar{x}\bar{y} + \bar{x}z) \cdot (xyz + x\bar{y}\bar{z} + \bar{x}\bar{y}\bar{z} + x\bar{y}\bar{z})$$

$$(\bar{x}\bar{y} \cdot \bar{x}z) \cdot (xyz + x\bar{y}\bar{z} + \bar{x}\bar{y}\bar{z} + x\bar{y}\bar{z})$$

$$[(\bar{x} + \bar{y}) \cdot (\bar{x} + \bar{z})] \cdot (xyz + x\bar{y}\bar{z} + \bar{x}\bar{y}\bar{z} + x\bar{y}\bar{z})$$

$$[(\bar{x} + \bar{y}) \cdot (\bar{x} + \bar{z})] \cdot (xyz + x\bar{y}\bar{z} + \bar{x}\bar{y}\bar{z} + x\bar{y}\bar{z})$$

$$(\bar{x}x + \bar{x}\bar{z} + \bar{y}x + \bar{y}\bar{z}) \cdot (xyz + x\bar{y}\bar{z} + \bar{x}\bar{y}\bar{z} + x\bar{y}\bar{z})$$

$$(0 + \bar{x}\bar{z} + \bar{y}x + \bar{y}\bar{z}) \cdot (xyz + x\bar{y}\bar{z} + \bar{x}\bar{y}\bar{z} + x\bar{y}\bar{z})$$

$$(\bar{x}\bar{z} + \bar{y}x + \bar{y}\bar{z}) \cdot (xyz + x\bar{y}\bar{z} + \bar{x}\bar{y}\bar{z} + x\bar{y}\bar{z})$$

$$(\bar{x}\bar{z} + \bar{y}x + \bar{y}\bar{z}) \cdot (xy(\bar{z} + \bar{z}) + \bar{x}\bar{y}\bar{z} + \bar{x}\bar{y}\bar{z})$$

$$(\bar{x}\bar{z} + \bar{y}x + \bar{y}\bar{z}) \cdot (xy(1) + \bar{x}\bar{y}\bar{z} + \bar{x}\bar{y}\bar{z})$$

$$(\bar{x}\bar{z} + \bar{y}x + \bar{y}\bar{z}) \cdot (xy + \bar{x}\bar{y}\bar{z} + \bar{x}\bar{y}\bar{z})$$

$$(\bar{x}\bar{z} + \bar{y}x + \bar{y}\bar{z}) \cdot (x(y + \bar{y}\bar{z}) + \bar{x}\bar{y}\bar{z})$$

$$(\bar{x}\bar{z} + \bar{y}x + \bar{y}\bar{z}) \cdot (x(y + \bar{z}) + \bar{x}\bar{y}\bar{z})$$

$$(\bar{x}\bar{z} + \bar{y}x + \bar{y}\bar{z}) \cdot (xy + x\bar{z} + \bar{x}\bar{y}\bar{z})$$

$$\bar{x}\bar{z}xy + \bar{x}\bar{z}x\bar{z} + \bar{x}\bar{z}\bar{x}\bar{y}\bar{z} + \bar{y}xxy + \bar{y}x\bar{y}\bar{z} + \bar{y}x\bar{y}\bar{z} + \bar{y}\bar{z}xy + \bar{y}\bar{z}x\bar{z} + \bar{y}\bar{z}\bar{x}\bar{y}\bar{z}$$

$$0 + 0 + \bar{x}\bar{y}\bar{z} + xy + x\bar{y}\bar{z} + 0 + x\bar{y}\bar{z} + 0 + 0$$

$$\bar{x}\bar{y}\bar{z} + xy + x\bar{y}\bar{z} + x\bar{y}\bar{z}$$

$$x \cdot (\bar{x}\bar{y}\bar{z} + xy + x\bar{y}\bar{z} + x\bar{y}\bar{z})$$

$$\bar{x}\bar{y}\bar{z} + xy + xy(1)$$

$$\bar{x}\bar{y}\bar{z} + xy + xy$$

$$\bar{x}\bar{y}\bar{z} + xy$$



$$(x \cdot \bar{y} + \bar{x} \cdot z)' \cdot (x \cdot y \cdot z + x \cdot \bar{y} \cdot \bar{z} + \bar{x} \cdot \bar{y} \cdot \bar{z} + x \cdot y \cdot \bar{z})$$

|   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 1 | 1 | 1 | 1 | 1 | 1 | 0 | 0 | 0 | 0 | 1 |
| 0 | 0 | 0 | 0 | 0 | 1 | 0 | 0 | 0 | 1 | 0 | 0 | 0 | 1 | 1 | 1 | 0 | 0 | 0 | 0 | 0 |
| 0 | 0 | 0 | 0 | 1 | 0 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 1 | 0 | 0 | 1 | 0 | 1 |
| 0 | 0 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 1 | 0 |
| 1 | 1 | 1 | 0 | 0 | 0 | 0 | 0 | 1 | 1 | 1 | 1 | 1 | 1 | 0 | 0 | 1 | 0 | 1 | 1 | 0 |
| 1 | 1 | 1 | 1 | 0 | 0 | 1 | 0 | 0 | 1 | 1 | 0 | 0 | 0 | 0 | 1 | 0 | 0 | 0 | 1 | 0 |
| 1 | 0 | 0 | 0 | 0 | 0 | 1 | 1 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 1 | 1 | 1 | 1 | 1 |
| 1 | 0 | 0 | 0 | 0 | 1 | 1 | 1 | 1 | 1 | 0 | 0 | 0 | 0 | 1 | 0 | 0 | 0 | 0 | 1 | 0 |

$$\bar{x} \cdot \bar{y} \cdot z + x \cdot y$$

|   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|
| 1 | 1 | 1 | 1 | 1 | 0 | 0 | 0 |
| 1 | 1 | 1 | 0 | 0 | 0 | 0 | 0 |
| 1 | 0 | 0 | 0 | 1 | 0 | 0 | 1 |
| 1 | 0 | 0 | 0 | 0 | 0 | 0 | 1 |
| 0 | 0 | 1 | 0 | 1 | 0 | 1 | 0 |
| 0 | 0 | 1 | 0 | 0 | 0 | 1 | 0 |
| 0 | 0 | 0 | 0 | 1 | 1 | 1 | 1 |
| 0 | 0 | 0 | 0 | 0 | 1 | 1 | 1 |

### Quinto Ejercicio

$$\begin{aligned} & ABCD + \bar{A}BCD + A\bar{B}CD + ABC\bar{D} + \bar{A}\bar{B}C\bar{D} + A\bar{B}\bar{C}\bar{D} + AB\bar{C}\bar{D} \\ & BCD(\bar{A} + A) + \bar{B}CD(A + \bar{A}) + A\bar{C}\bar{D}(B + \bar{B}) + ABC\bar{D} \\ & BCD(1) + \bar{B}CD(1) + A\bar{C}\bar{D}(1) + ABC\bar{D} \\ & BCD + \bar{B}CD + A\bar{C}\bar{D} + ABC\bar{D} \\ & CD(B + \bar{B}) + A\bar{C}\bar{D} + ABC\bar{D} \\ & CD(1) + A\bar{C}\bar{D} + ABC\bar{D} \\ & CD + A\bar{C}\bar{D} + ABC\bar{D} \\ & D(C + A\bar{C}) + ABC\bar{D} \\ & D(C + A) + ABC\bar{D} \\ & CD + AD + ABC\bar{D} \\ & CD + A(D + BC\bar{D}) \\ & CD + A(D + BC) \\ & CD + AD + ABC \end{aligned}$$



[illegible]

| D | C | A | D | A | B | C |
|---|---|---|---|---|---|---|
| 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 1 | 0 | 0 | 0 | 0 | 0 | 0 |
| 0 | 0 | 1 | 0 | 0 | 0 | 0 |
| 1 | 1 | 1 | 1 | 0 | 0 | 1 |
| 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 1 | 0 | 0 | 0 | 0 | 1 | 0 |
| 0 | 0 | 1 | 0 | 0 | 0 | 0 |
| 1 | 1 | 1 | 1 | 0 | 0 | 1 |
| 0 | 0 | 0 | 0 | 1 | 0 | 0 |
| 1 | 0 | 0 | 1 | 0 | 0 | 0 |
| 0 | 0 | 1 | 0 | 0 | 0 | 0 |
| 1 | 1 | 1 | 1 | 1 | 1 | 1 |
| 0 | 0 | 1 | 0 | 0 | 0 | 0 |
| 1 | 0 | 0 | 1 | 1 | 1 | 1 |
| 0 | 0 | 1 | 0 | 0 | 0 | 0 |
| 1 | 1 | 1 | 1 | 1 | 1 | 1 |
| 0 | 0 | 0 | 0 | 1 | 1 | 0 |
| 1 | 0 | 0 | 1 | 1 | 1 | 0 |
| 0 | 0 | 1 | 0 | 0 | 1 | 1 |
| 1 | 1 | 1 | 1 | 1 | 1 | 1 |